TASK 3 – Choose Variables and Name the Constants

Naming conventions

Use descriptive snake_case names for variables and functions, such as farmer_id, fat_percent and rate_per_litre. Use uppercase names with underscores for constants: BASE_RATE, REFERENCE_FAT, REFERENCE_SNF, FAT_STEP_VALUE, SNF_STEP_VALUE and STEP_SIZE. Names should describe purpose rather than type and should not shadow built-in names such as list, sum or input.

Constants and dynamic typing

Python does not enforce immutable constants. Uppercase is a convention that tells developers not to reassign a value. Keeping the rate card in one place makes seasonal or policy changes safer. Python is dynamically typed: a name refers to an object and can be rebound to a different type. For example, value = 10 makes value refer to an int; value = "ten" later makes it refer to a str. Multiple assignment evaluates the right-hand side before assigning: a, b = b, a swaps the two values without a temporary variable.

Recommended constants and example

The following design centralises the original rate card. Decimal is preferred for currency values.

Code

```
from decimal import Decimal

BASE_RATE = Decimal("30.00")
REFERENCE_FAT = Decimal("3.5")
REFERENCE_SNF = Decimal("8.5")
FAT_STEP_VALUE = Decimal("0.40")
SNF_STEP_VALUE = Decimal("0.30")
STEP_SIZE = Decimal("0.1")
MIN_FAT = Decimal("2.0")
MAX_FAT = Decimal("7.0")

first, second = 10, 20
first, second = second, first
print(first, second)  # 20 10
```

Alternative and complexity

Magic numbers typed directly into expressions make a rate revision a search-and-edit operation across k locations, creating a risk that one occurrence is missed. Named constants require one update. A dictionary such as RATE_CARD = {"base": 30.0, "fat": 0.40, "snf": 0.30} also centralises values and is useful when selecting among multiple rate cards. Accessing a constant or dictionary key is O(1) on average. Runtime differences are negligible for this application; the principal difference is maintainability. A dictionary has a small additional memory cost for its keys and values.